BACKPLANE / CONNECTION BLOCKS

PCB-oriented top view | Continuous wires | Test points omitted | 21 Oct 2026

U2D2 / DYNAMIXEL

M1 / U2D2

USB to host computer

Mount: no PCB signal pads

J7 / SERVO BUS

DYNAMIXEL_DATA

+12V

GND

ARDUINO MEGA / TEENSY ADAPTER

UNUSED CONTACTS / NC

A1 / MEGA-PATTERN SOCKET

Alternative: Teensy on adapter

FEMALE_1_/_LIGHT_SENSOR
FEMALE_2_/_LIGHT_SENSOR
FEMALE_3_/_LIGHT_SENSOR
MALE_1_/_LIGHT_SENSOR_/_B
MALE_1_/_LIGHT_SENSOR_/_D
MALE_1_/_LIGHT_SENSOR_/_A
MALE_1_/_LIGHT_SENSOR_/_C
MALE_2_/_LIGHT_SENSOR_/_B
MALE_2_/_LIGHT_SENSOR_/_D
MALE_2_/_LIGHT_SENSOR_/_A
MALE_2_/_LIGHT_SENSOR_/_C
FEMALE_1_/_ANALYSER_OUT
FEMALE_2_/_ANALYSER_OUT
FEMALE_3_/_ANALYSER_OUT
MALE_1_/_ANALYSER_OUT
MALE_2_/_ANALYSER_OUT
FEMALE_1_/_MIC_DIRECT
FEMALE_2_/_MIC_DIRECT
FEMALE_3_/_MIC_DIRECT
MALE_1_/_MIC_DIRECT
MALE_2_/_MIC_DIRECT

FEMALE_1_/_TONE
FEMALE_2_/_TONE
FEMALE_3_/_TONE
MALE_1_/_TONE
MALE_2_/_TONE
ANALYSER_/_STROBE
ANALYSER_/_RESET
FEMALE_1_/_NEOPIXEL_DATA_/_DRIVEN
FEMALE_2_/_NEOPIXEL_DATA_/_DRIVEN
FEMALE_3_/_NEOPIXEL_DATA_/_DRIVEN
MALE_1_/_NEOPIXEL_DATA_/_DRIVEN
MALE_2_/_NEOPIXEL_DATA_/_DRIVEN
MALE_1_/_BAR_NEOPIXEL_DATA_/_DRIVEN
MALE_2_/_BAR_NEOPIXEL_DATA_/_DRIVEN
AMP_SHUTDOWN
MALE_1_/_AUX_DRIVEN
MALE_2_/_AUX_DRIVEN

GND + MEGA_5V contacts

POWER ENTRY / SUPPLIES

J2 / 5 V INPUT

+5V
GND

J6 / SCREW BRIDGE

GND
GND
+5V
GND
+5V
+5V

C1 470uF 25V

C2 100nF

JP1 OR

AGND / GND bond

RLP5 2K2

RLP12 4K7

RLPUSB 2K2

DP5

DP12

DPUSB

LIGHT SENSOR LOADS

RP1 10K provisional
RP2 10K provisional
RP3 10K provisional
RP4 10K provisional
RP5 10K provisional
RP6 10K provisional
RP7 10K provisional
RP8 10K provisional
RP9 10K provisional
RP10 10K provisional
RP11 10K provisional

LED DATA / AUX OUTPUTS

RS1 10K

AMP SHUTDOWN PULLUP

RN1 330R
RN2 330R
RN3 330R
RN4 330R
RN5 330R
RN6 330R
RN7 330R

RND1 100K
RND2 100K
RND3 100K
RND4 100K
RND5 100K
RND6 100K
RND7 100K

RA1 160R

RA2 160R

J1 / FEMALE 2

0	GND
1	FEMALE 2 / SPARE1
2	FEMALE 2 / SPARE2
3	FEMALE 2 / SPARE3
4	FEMALE 2 / AUDIO RETURN
5	FEMALE 2 / LIGHT SENSOR
6	FEMALE 2 / MIC
7	+12V
8	GND
9	FEMALE 2 / SPARE4
10	FEMALE 2 / SPARE5
11	FEMALE 2 / SPARE6
12	FEMALE 2 / LINE OUT
13	FEMALE 2 / NEOPIXEL DATA
14	DYNAMIXEL DATA
15	+5V

J5 / FEMALE 1

0	GND
1	FEMALE 1 / SPARE1
2	FEMALE 1 / SPARE2
3	FEMALE 1 / SPARE3
4	FEMALE 1 / AUDIO RETURN
5	FEMALE 1 / LIGHT SENSOR
6	FEMALE 1 / MIC
7	+12V
8	GND
9	FEMALE 1 / SPARE4
10	FEMALE 1 / SPARE5
11	FEMALE 1 / SPARE6
12	FEMALE 1 / LINE OUT
13	FEMALE 1 / NEOPIXEL DATA
14	DYNAMIXEL DATA
15	+5V

A-J3 / FEMALE 3 + MALE 1

0	GND
1	GND
2	+12V
3	FEMALE 3 / MIC
4	FEMALE 3 / LIGHT SENSOR
5	FEMALE 3 / AUDIO RETURN
6	MALE 1 / NEOPIXEL DATA
7	MALE 1 / LIGHT SENSOR / B
8	MALE 1 / LIGHT SENSOR / D
9	+5V
10	DYNAMIXEL DATA
11	FEMALE 3 / NEOPIXEL DATA
12	FEMALE 3 / LINE OUT
13	MALE 1 / MIC
14	MALE 1 / LIGHT SENSOR / A
15	MALE 1 / LIGHT SENSOR / C

B-J4 / MALES + BAR

0	GND
1	MALE 1 / AUX
2	MALE 1 / AUDIO RETURN
3	MALE 2 / NEOPIXEL DATA
4	MALE 2 / LIGHT SENSOR / B
5	MALE 2 / LIGHT SENSOR / D
6	MALE 2 / LINE OUT
7	CENTRE / SPARE1
8	MALE 2 / BAR NEOPIXEL DATA
9	MALE 1 / LINE OUT
10	MALE 2 / MIC
11	MALE 2 / LIGHT SENSOR / A
12	MALE 2 / LIGHT SENSOR / C
13	MALE 2 / AUX
14	MALE 2 / AUDIO RETURN
15	MALE 1 / BAR NEOPIXEL DATA

VOICES / FIVE CARD SOCKETS

FEMALE 1 — JV1
FEMALE 2 — JV2
FEMALE 3 — JV3
MALE 1 — JV4
MALE 2 — JV5

+5V
TONE
AUDIO_OUT
GND
AGND

EXTRA1 / F2 SPARES

FEMALE_2_/_SPARE1
FEMALE_2_/_SPARE2
FEMALE_2_/_SPARE3
FEMALE_2_/_SPARE4
FEMALE_2_/_SPARE5
FEMALE_2_/_SPARE6

RLT1 1K5
RLT2 1K5
RLT3 1K5
RLT4 1K5
RLT5 1K5
R203 100R

DT1
DT2
DT3
DT4
DT5

R303 100R
R403 100R
R503 100R
R103 100R

RL1 100K
RL2 100K
RL3 100K
RL4 100K
RL5 100K

JP4 0R
JP5 0R
JP6 0R
JP2 0R
JP3 0R

HV1
HV2
HV3
HV4
HV5

EXTRA2 / F1 SPARES

FEMALE_1_/_SPARE1
FEMALE_1_/_SPARE2
FEMALE_1_/_SPARE3
FEMALE_1_/_SPARE4
FEMALE_1_/_SPARE5
FEMALE_1_/_SPARE6

AUDIO ANALYSERS / FIVE CHANNELS

JA1

DFR0126 ANALYSER CARRIER

5 microphones -> 5 analogue outputs

Shared RESET / STROBE

MEGA_5V supply / AGND returns

FEMALE_1_/_MIC
AGND
FEMALE_1_/_ANALYSER_OUT
FEMALE_2_/_MIC
AGND
FEMALE_2_/_ANALYSER_OUT
FEMALE_3_/_MIC
AGND
FEMALE_3_/_ANALYSER_OUT
MALE_1_/_MIC
AGND
MALE_1_/_ANALYSER_OUT
MALE_2_/_MIC
AGND
MALE_2_/_ANALYSER_OUT
AGND
ANALYSER_/_STROBE
ANALYSER_/_RESET
MEGA_5V
GND
GND

HA1
HA2

DIRECT MICROPHONE INPUTS

FEMALE 1 — RM1 4K7 — RMD1 1M
FEMALE 2 — RM2 4K7 — RMD2 1M
FEMALE 3 — RM3 4K7 — RMD3 1M
MALE 1 — RM4 4K7 — RMD4 1M
MALE 2 — RM5 4K7 — RMD5 1M

CENTRE SPARE

CENTRE_/_SPARE1

FEMALE 2 HARNESS
J1 / DSUB-15

FEMALE 1 HARNESS
J5 / DSUB-15

A-J3 / DSUB-15
FEMALE 3 / MALE 1 HARNESS

B-J4 / DSUB-15
MALE 1 / MALE 2 / BAR HARNESS

GND
AGND
+5V
+12V
MEGA_5V

AGND = ANALOG RETURN | MEGA_5V = CONTROLLER 5 V | Dots join wires; crossings without dots stay separate.